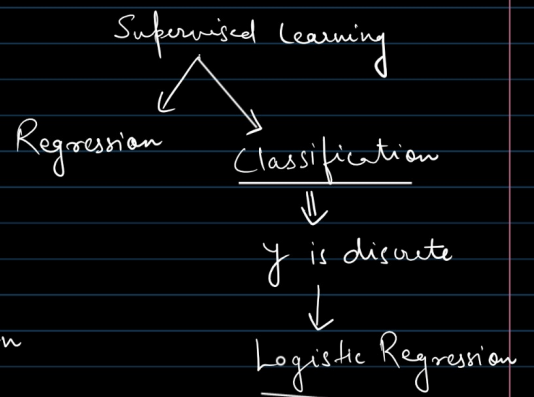
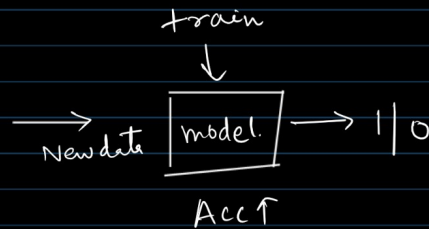


* Logistic Regression

eg. # of hours studied $\rightarrow$ Predict pass or fail.

X (# of hours)	Y Pass / Fail
2	0
8	1
3	0
4	0
6	1



eg. Predict if a person will fraud / Not

Salary (k)	Age (years)	Fraud / No fraud
20	19	1
22.5	20	1
30	35	0
-	-	-

train $\rightarrow$ model $\rightarrow$ 1/0

50k, 30 $\rightarrow$

eg. Predict diabetes / not diabetes

FBS level	diabetes (1/0)
100	1
250	1
80	0
70	0

80 $\rightarrow$ model $\rightarrow$ 1/0

eg. Cancer / Not Cancer based on size of tumor.

Size of tumor (mm)	Cancer / Not Cancer
1.2	0
3.2	1
4.1	1
5.6	1
-	-
-	-

eg Spam | ham

Classification

bi-2 binary classification
(output: 2 categories)

pass/fail, Diabetic/Not diabetic,
Cancer/Not Cancer.

> multiclass classification

(output: more than 2 categories)

eg. Good, fair bad.

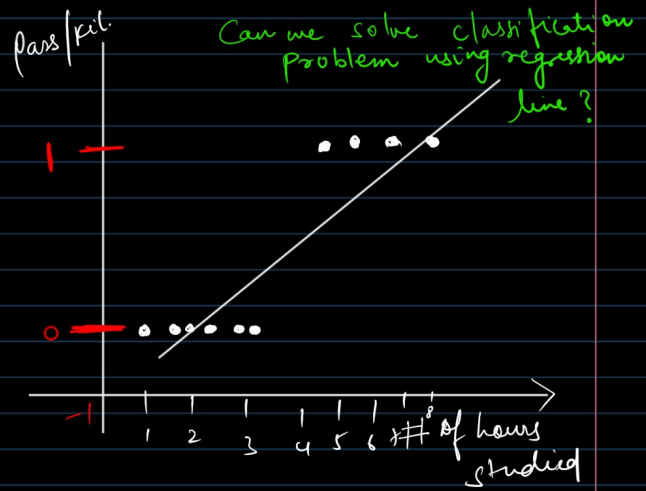
Age	Salary	<u>Credit score</u>
20	50k	Good ————— 2
25	30k	Fair ————— 1
20	31k	Good ————— 2
30	50k	bad ————— 0

multiclass
↓
many

→ There are different Algorithms to solve Classification Problem.

* Logistic Regression

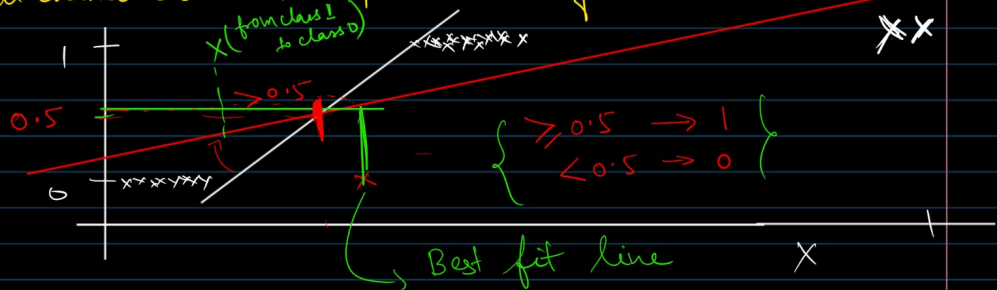
X (No. of hours)	Pass/Fail
2	0
8	1
3	0
4	0
6	1
-	-
-	-



* Why we cannot use regression line for classification problem?

→ Nature of Problem statement is different: → In regression the target variable y was continuous in nature, but here its discrete value.

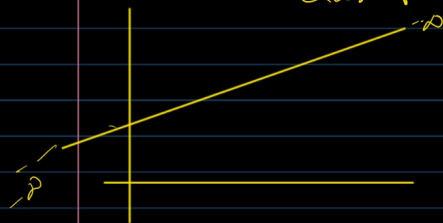
→ Best fit line will change due to presence of outlier.



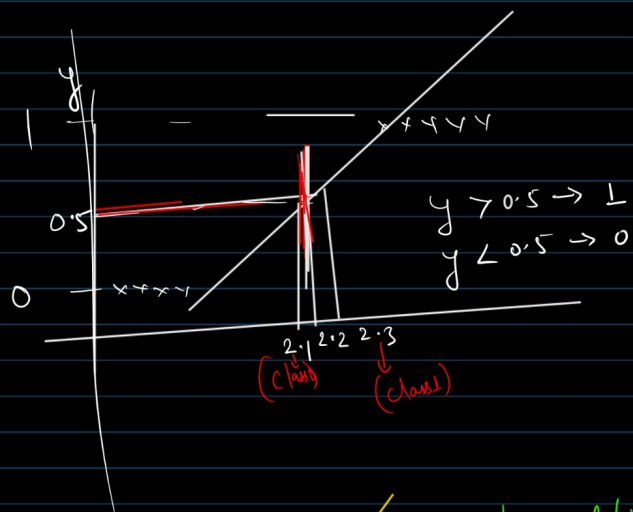
→ Related to first point:

Regression line range: $-\infty$ to ∞

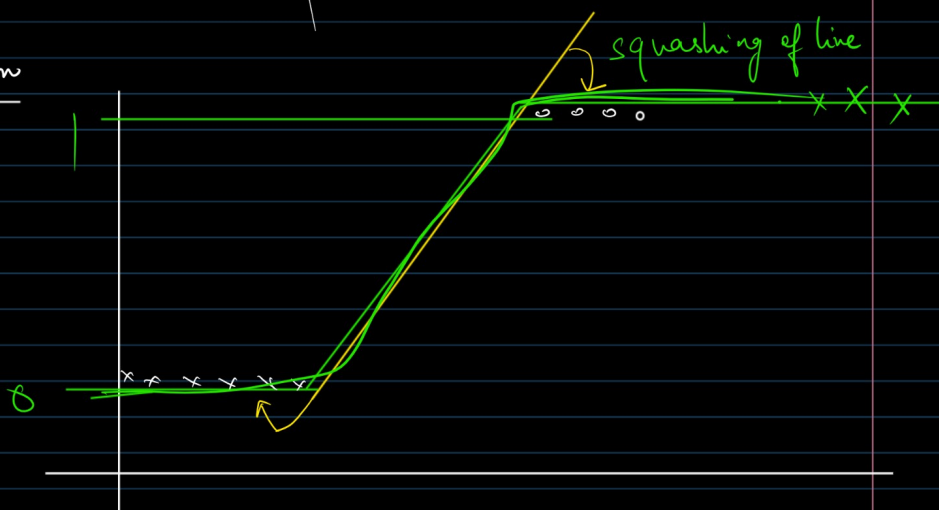
Classification problem target: 0 to 1 range



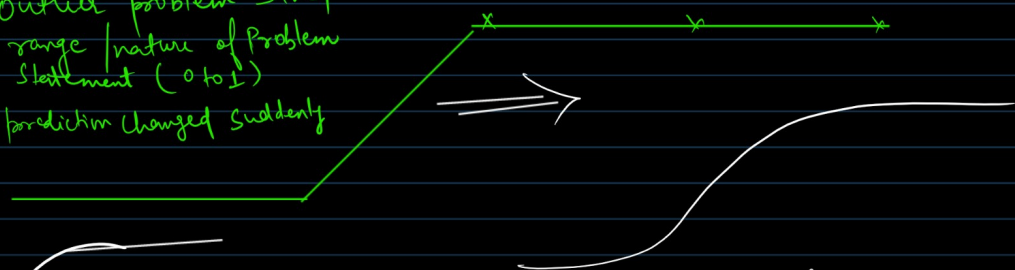
→ Prediction changes suddenly
↳ decision boundary
is not
changing
gradually.



* How to solve these problem



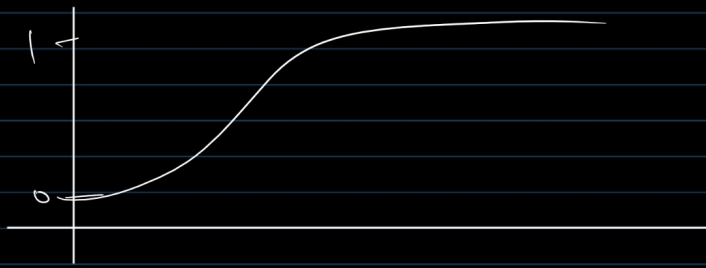
- ✓ → outlier problem solved
- ✓ → range / nature of Problem statement (0 to 1)
- ✓ → prediction changed suddenly



→ Sigmoid function (many Sigmoid fn)
↓
logistic function (specific for logistic reg model)

$$\sigma = \frac{1}{1+e^{-z}}$$

0 to 1



$\sigma = \frac{1}{1+e^{-z}}$

$-\infty$ to ∞ $\longrightarrow$ 0 to 1

$z = -\infty \Rightarrow \frac{1}{1+e^{-(-\infty)}} = \frac{1}{1+e^{\infty}} = \frac{1}{1+\infty} = \frac{1}{\infty} = 0$

$z = \infty \Rightarrow \frac{1}{1+e^{-\infty}} \Rightarrow \frac{1}{1+\frac{1}{e^{\infty}}} = \frac{1}{1+\frac{1}{\infty}} = \frac{1}{1+0} = 1$

$\frac{1}{3} = \frac{1}{3}$

$h_{\theta}(x) = \theta_0 + \theta_1 x_1 \longrightarrow$ Best fit line.

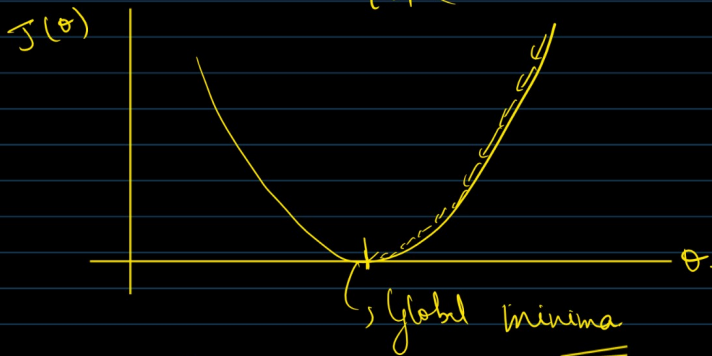
$$h_{\theta}(x) = \frac{1}{1+e^{-(\theta_0 + \theta_1 x_1)}}$$

$\longrightarrow$ Logistic regression model.

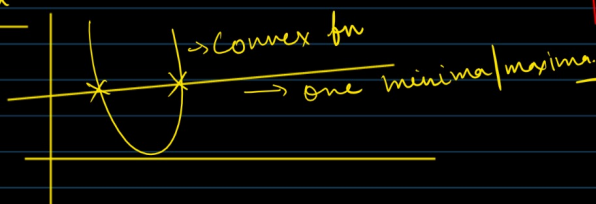
$\longrightarrow$ To get optimal θ_0 & θ_1 , minimise the Cost function.

Linear Regression model

$$J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - h_{\theta}(x_i))^2$$



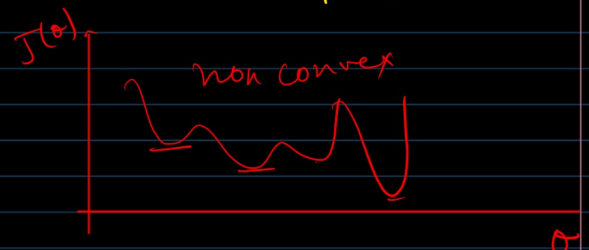
* Convex fn



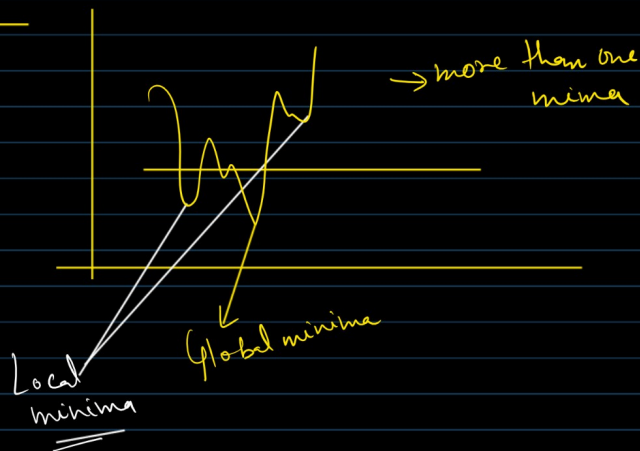
Logistic Regression

$$J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \underbrace{h_{\theta}(x_i)}_{\frac{1}{1+e^{-(\theta_0 + \theta_1 x_1)}}})^2$$

$$h_{\theta}(x) = \frac{1}{1+e^{-(\theta_0 + \theta_1 x_1)}}$$



* non Convex



* In ML algorithm, the Convex Cost fn is preferred. but original cost fn for logistic regression model is non convex. So it is tweaked a little bit to get a new CF.

log loss function

$$J(\theta_0, \theta_1) = -y_i \log(h_0(x_i)) - (1-y_i) \log(1-h_0(x_i))$$

$$\text{where } h_0(x_i) = \frac{1}{1 + e^{-\theta_0 + \theta_1 x}}$$

y_i — actual value

$$J(\theta_0, \theta_1) = -y_i \log h_0(x_i) - (1-y_i) \log(1-h_0(x_i))$$

$\downarrow y=0 \Rightarrow 0$ $\downarrow y=1 \Rightarrow 0$

$$J(\theta_0, \theta_1) \begin{cases} = -\log h_0(x_i) & y = 1 \\ = -\log(1-h_0(x_i)) & y = 0 \end{cases}$$

To minimise the cost function $J(\theta_0, \theta_1)$ by changing θ_0 & θ_1

Convergence Algorithm

Repeat until convergence

$$\left\{ \theta_j : \theta_j = \theta_j - \frac{\partial J(\theta_0, \theta_1)}{\partial \theta} \right\}$$

One IV

To get optimal θ_0, θ_1

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1)}}$$

multiple IV

↓
multivariate
Logistic
Regression

$$h_{\theta}(x) = \frac{1}{1 + e^{-(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n)}}$$

Logistic function $\rightarrow$ 0 to 1

Output of logistic Regression model:

↓
0 to 1

